

Air University



Network Security (Lab)

Year 2026

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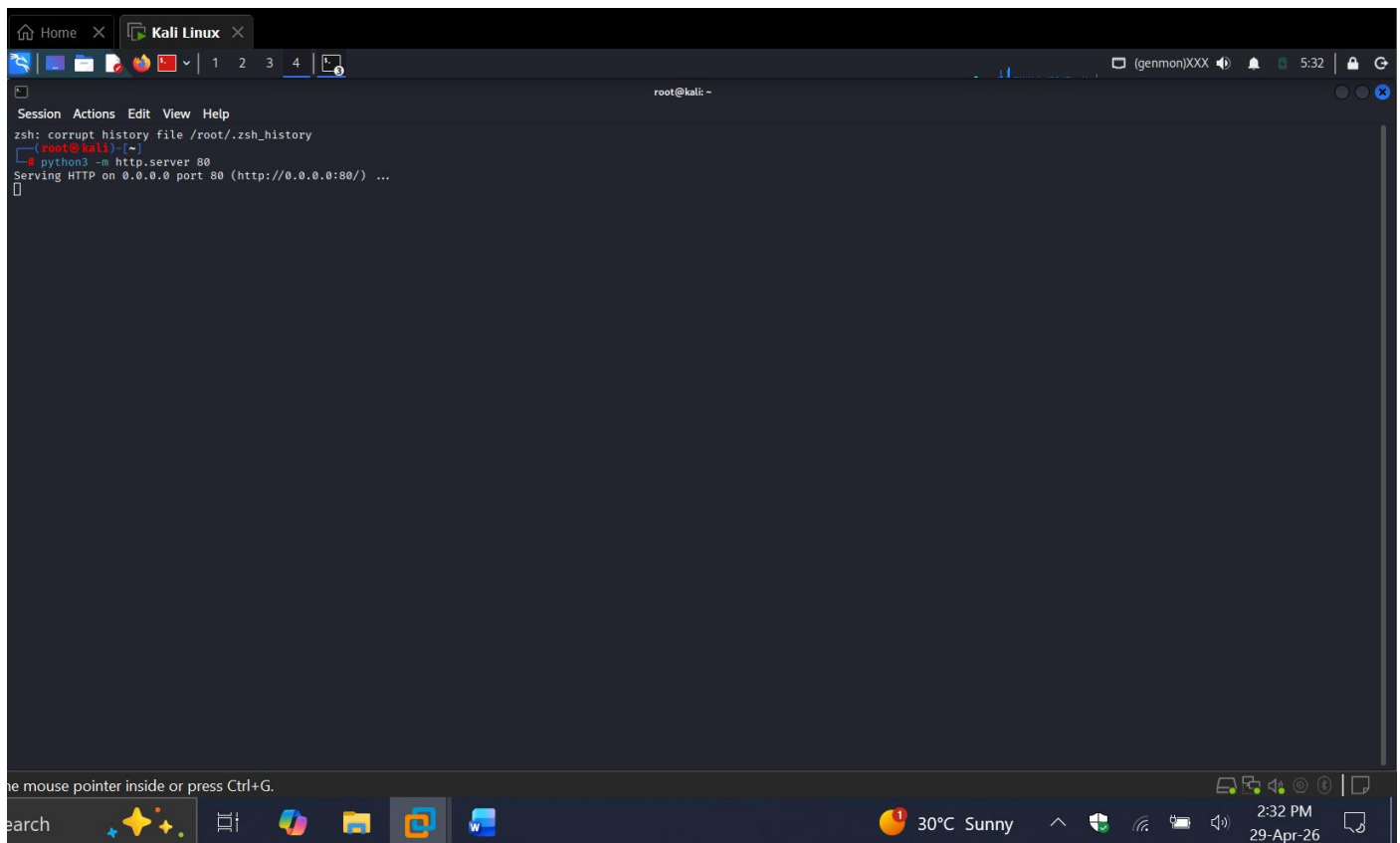
Academic Year: 2024 – 2028
Department of Cyber Security

USAMA ARSHAD | 247141
BS – Cyber Security – Fall – 24 (A)
Date of Submission: April 29, 2026

Lab # 13

Task 1: TCP Sync Flood Attack

Initializing the server:

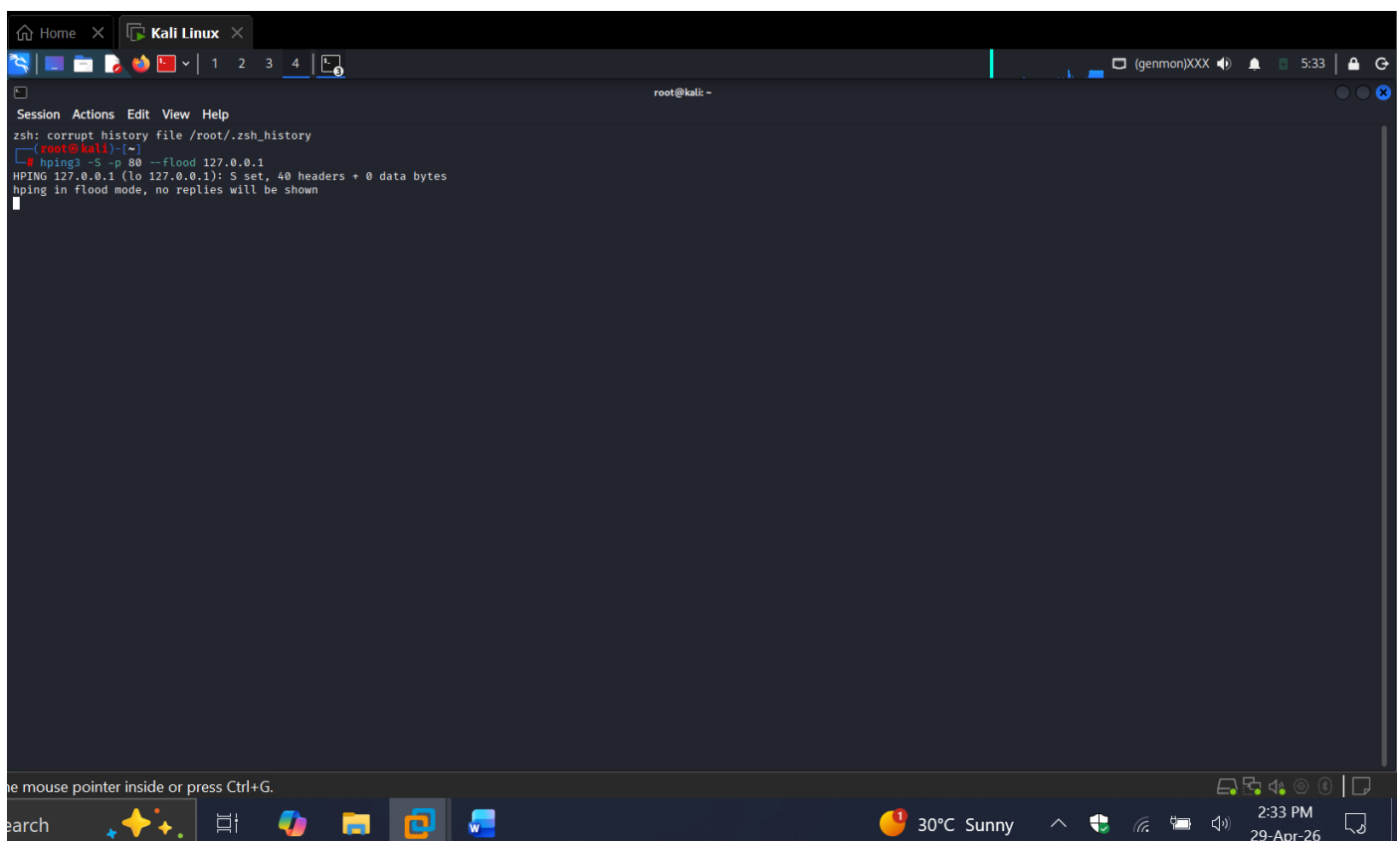


The screenshot shows a Kali Linux terminal window with the following text:

```
Session Actions Edit View Help
zsh: corrupt history file /root/.zsh_history
root@kali: ~
python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
```

The terminal window is titled "Kali Linux" and shows the user is root. The command prompt is root@kali: ~. The user has run python3 -m http.server 80, and the output is Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...

Launch TCP sync flood:



The screenshot shows a Kali Linux terminal window with the following text:

```
Session Actions Edit View Help
zsh: corrupt history file /root/.zsh_history
root@kali: ~
hping3 -S -p 80 --flood 127.0.0.1
HPING 127.0.0.1 (lo 127.0.0.1): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

The terminal window is titled "Kali Linux" and shows the user is root. The command prompt is root@kali: ~. The user has run hping3 -S -p 80 --flood 127.0.0.1, and the output is HPING 127.0.0.1 (lo 127.0.0.1): S set, 40 headers + 0 data bytes. The user has then run hping in flood mode, no replies will be shown.

From Wireshark:

Wireshark packet capture showing a series of TCP RST and SYN packets between 127.0.0.1 and 127.0.0.1. The capture is on the 'Loopback: lo' interface. The packet list shows packets 829297 through 829328. The packet details pane shows the selected packet (829328) as a Transmission Control Protocol packet, Src Port: 16721, Dst Port: 80, Seq: 0, Len: 0. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
829297	3.021198337	127.0.0.1	127.0.0.1	TCP	58	80 → 31011 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829298	3.021200719	127.0.0.1	127.0.0.1	TCP	54	31011 → 80 [RST] Seq=1 Win=0 Len=0
829299	3.021208368	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31012 → 80 [SYN] Seq=0 Win=512 Len=0
829300	3.021214197	127.0.0.1	127.0.0.1	TCP	58	80 → 31012 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829301	3.021216134	127.0.0.1	127.0.0.1	TCP	54	31012 → 80 [RST] Seq=1 Win=0 Len=0
829302	3.021221231	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31013 → 80 [SYN] Seq=0 Win=512 Len=0
829303	3.021224752	127.0.0.1	127.0.0.1	TCP	58	80 → 31013 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829304	3.021228902	127.0.0.1	127.0.0.1	TCP	54	31013 → 80 [RST] Seq=1 Win=0 Len=0
829305	3.021238375	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31014 → 80 [SYN] Seq=0 Win=512 Len=0
829306	3.021233644	127.0.0.1	127.0.0.1	TCP	58	80 → 31014 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829307	3.021235242	127.0.0.1	127.0.0.1	TCP	54	31014 → 80 [RST] Seq=1 Win=0 Len=0
829308	3.021239938	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31015 → 80 [SYN] Seq=0 Win=512 Len=0
829309	3.021243319	127.0.0.1	127.0.0.1	TCP	58	80 → 31015 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829310	3.021244984	127.0.0.1	127.0.0.1	TCP	54	31015 → 80 [RST] Seq=1 Win=0 Len=0
829311	3.021249234	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31016 → 80 [SYN] Seq=0 Win=512 Len=0
829312	3.021252548	127.0.0.1	127.0.0.1	TCP	58	80 → 31016 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829313	3.021254279	127.0.0.1	127.0.0.1	TCP	54	31016 → 80 [RST] Seq=1 Win=0 Len=0
829314	3.021258610	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31017 → 80 [SYN] Seq=0 Win=512 Len=0
829315	3.021260916	127.0.0.1	127.0.0.1	TCP	58	80 → 31017 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829316	3.021263753	127.0.0.1	127.0.0.1	TCP	54	31017 → 80 [RST] Seq=1 Win=0 Len=0
829317	3.021267897	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31018 → 80 [SYN] Seq=0 Win=512 Len=0
829318	3.021271115	127.0.0.1	127.0.0.1	TCP	58	80 → 31018 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829319	3.021272740	127.0.0.1	127.0.0.1	TCP	54	31018 → 80 [RST] Seq=1 Win=0 Len=0
829320	3.021276746	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31019 → 80 [SYN] Seq=0 Win=512 Len=0
829321	3.021279908	127.0.0.1	127.0.0.1	TCP	58	80 → 31019 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829322	3.021281467	127.0.0.1	127.0.0.1	TCP	54	31019 → 80 [RST] Seq=1 Win=0 Len=0
829323	3.021285773	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31020 → 80 [SYN] Seq=0 Win=512 Len=0
829324	3.021289218	127.0.0.1	127.0.0.1	TCP	58	80 → 31020 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829325	3.021290916	127.0.0.1	127.0.0.1	TCP	54	31020 → 80 [RST] Seq=1 Win=0 Len=0
829326	3.021295129	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 31021 → 80 [SYN] Seq=0 Win=512 Len=0
829327	3.021298556	127.0.0.1	127.0.0.1	TCP	58	80 → 31021 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
829328	3.021322806	127.0.0.1	127.0.0.1	TCP	54	31021 → 80 [RST] Seq=1 Win=0 Len=0

Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 'Loopback: lo'. Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00). Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1. Transmission Control Protocol, Src Port: 16721, Dst Port: 80, Seq: 0, Len: 0.

The Kali VM has crashed:

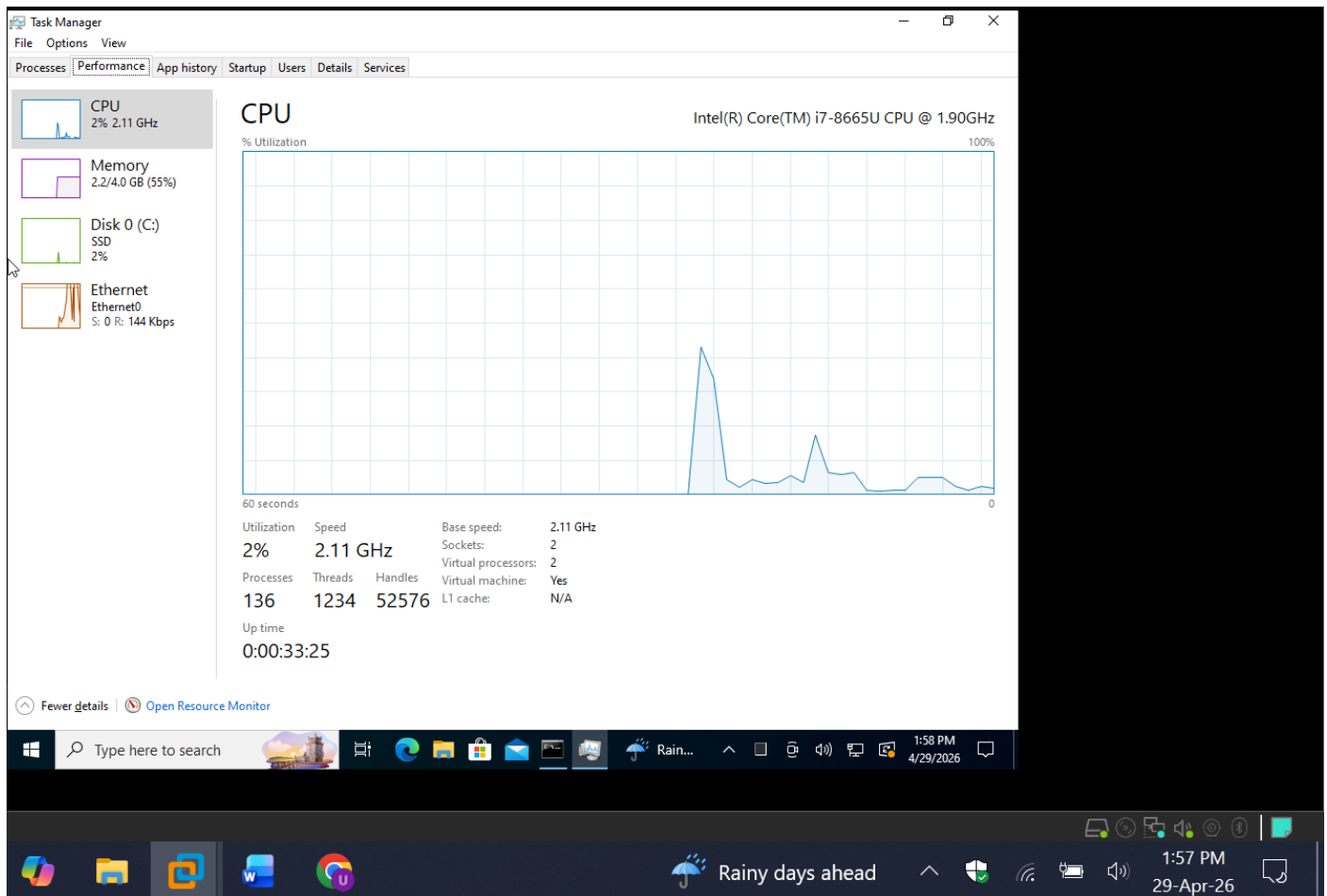
Wireshark packet capture showing a series of TCP RST and SYN packets between 127.0.0.1 and 127.0.0.1. The capture is on the 'Loopback: lo' interface. The packet list shows packets 422600 through 422606. The packet details pane shows the selected packet (422606) as a Transmission Control Protocol packet, Src Port: 16721, Dst Port: 80, Seq: 0, Len: 0. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
422600	16.216719935	127.0.0.1	127.0.0.1	TCP	54	80 → 49138 [RST] Seq=1 Win=0 Len=0
422601	16.216723445	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49139 → 80 [SYN] Seq=0 Win=512 Len=0
422602	16.216726094	127.0.0.1	127.0.0.1	TCP	58	80 → 49139 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422603	16.216727418	127.0.0.1	127.0.0.1	TCP	54	49139 → 80 [RST] Seq=1 Win=0 Len=0
422604	16.216730901	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49140 → 80 [SYN] Seq=0 Win=512 Len=0
422605	16.216733588	127.0.0.1	127.0.0.1	TCP	58	80 → 49140 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422606	16.216734957	127.0.0.1	127.0.0.1	TCP	54	49140 → 80 [RST] Seq=1 Win=0 Len=0
422607	16.216738636	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49141 → 80 [SYN] Seq=0 Win=512 Len=0
422608	16.216741590	127.0.0.1	127.0.0.1	TCP	58	80 → 49141 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422609	16.216742963	127.0.0.1	127.0.0.1	TCP	54	49141 → 80 [RST] Seq=1 Win=0 Len=0
422610	16.216746819	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49142 → 80 [SYN] Seq=0 Win=512 Len=0
422611	16.216749945	127.0.0.1	127.0.0.1	TCP	58	80 → 49142 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422612	16.216751319	127.0.0.1	127.0.0.1	TCP	54	49142 → 80 [RST] Seq=1 Win=0 Len=0
422613	16.216754871	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49143 → 80 [SYN] Seq=0 Win=512 Len=0
422614	16.216757722	127.0.0.1	127.0.0.1	TCP	58	80 → 49143 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422615	16.216759089	127.0.0.1	127.0.0.1	TCP	54	49143 → 80 [RST] Seq=1 Win=0 Len=0
422616	16.216762726	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49144 → 80 [SYN] Seq=0 Win=512 Len=0
422617	16.216765925	127.0.0.1	127.0.0.1	TCP	58	80 → 49144 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422618	16.216767335	127.0.0.1	127.0.0.1	TCP	54	49144 → 80 [RST] Seq=1 Win=0 Len=0
422619	16.216771491	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49145 → 80 [SYN] Seq=0 Win=512 Len=0
422620	16.216774235	127.0.0.1	127.0.0.1	TCP	58	80 → 49145 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422621	16.216776604	127.0.0.1	127.0.0.1	TCP	54	49145 → 80 [RST] Seq=1 Win=0 Len=0
422622	16.216779364	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49146 → 80 [SYN] Seq=0 Win=512 Len=0
422623	16.216782310	127.0.0.1	127.0.0.1	TCP	58	80 → 49146 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422624	16.216783674	127.0.0.1	127.0.0.1	TCP	54	49146 → 80 [RST] Seq=1 Win=0 Len=0
422625	16.216787252	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49147 → 80 [SYN] Seq=0 Win=512 Len=0
422626	16.216789988	127.0.0.1	127.0.0.1	TCP	58	80 → 49147 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422627	16.216791288	127.0.0.1	127.0.0.1	TCP	54	49147 → 80 [RST] Seq=1 Win=0 Len=0
422628	16.216794793	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49148 → 80 [SYN] Seq=0 Win=512 Len=0
422629	16.216797506	127.0.0.1	127.0.0.1	TCP	58	80 → 49148 [SYN, ACK] Seq=0 Ack=1 Win=65495 Len=0 MSS=65495
422630	16.216798855	127.0.0.1	127.0.0.1	TCP	54	49148 → 80 [RST] Seq=1 Win=0 Len=0
422631	16.216819650	127.0.0.1	127.0.0.1	TCP	54	[TCP Port numbers reused] 49149 → 80 [SYN] Seq=0 Win=512 Len=0

Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 'Loopback: lo'. Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00). Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1. Transmission Control Protocol, Src Port: 16721, Dst Port: 80, Seq: 0, Len: 0.

Task 2: ICMP Flood Attack

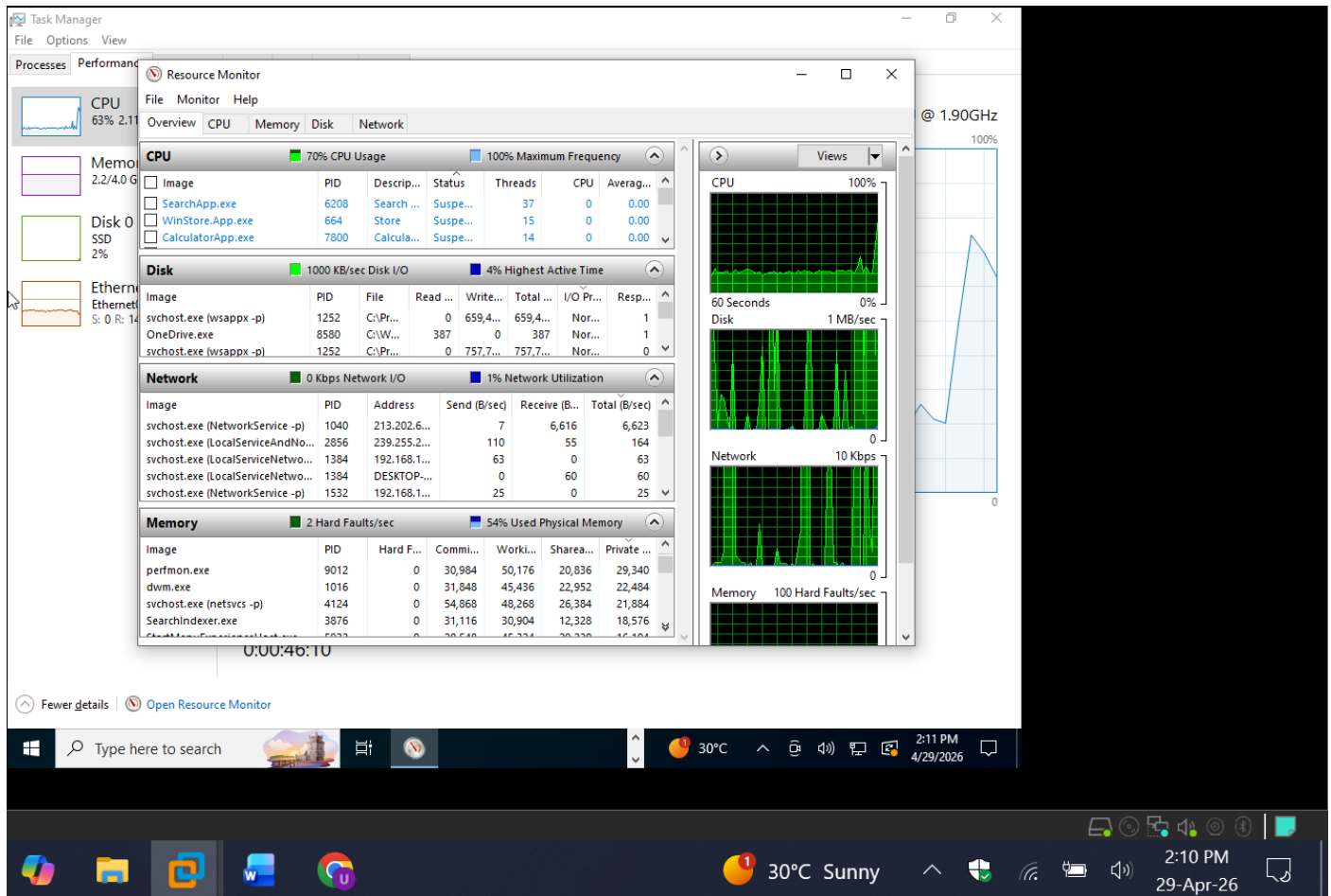
Before Attack:



Start the ICMP flood:

```
(root@kali)~# hping3 --icmp --flood 192.168.180.134
HPING 192.168.180.134 (eth0 192.168.180.134): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

After Attack:



From Wireshark:

